Unsharp masking

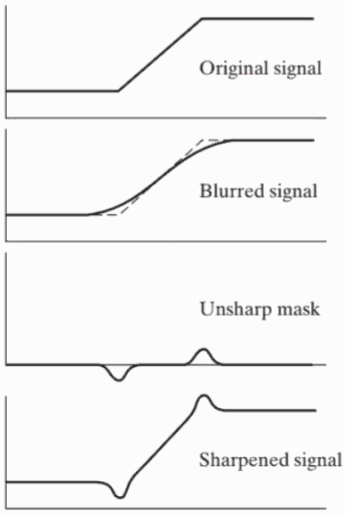
$f_s(x,y) = f(x,y) - \bar{f}(x,y)$ ← bulanıklaştırılmış fotoğraf

f_s → mask

→ keskinleştirilmiş maske

$g(x,y) = f(x,y) + f_s(x,y)$

$= f(x,y) + f(x,y) - \bar{f}(x,y)$



a
b
c
d

FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

DIP-XE → $f(x,y)$

DIP-XE $\bar{f}(x,y)$

DIP-XE f_s göstermek için yapmış

DIP-XE → $g(x,y)$ Unsharp mask

DIP-XE

a
b
c
d
e

FIGURE 3.40 (a) Original image. (b) Result of blurring with a Gaussian filter. (c) Unsharp mask. (d) Result of using unsharp masking. (e) Result of using highboost filtering.

$f_{hb} = A f(x,y) - \bar{f}(x,y)$ ← $A > 1$ ← genelde

kitap

$f_{Hb} = f(x,y) + k f_s(x,y)$

$k > 0$

a
b c
d e

FIGURE 3.41
A 3 × 3 region of an image (the zs are intensity values). (b)–(c) Roberts cross gradient operators. (d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

0	−1
1	0

Sobel

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

1
0
-1

dike yönde x'e göre türev

1	0	-1

1/3	1/3	1/3
0	0	0
-1/3	-1/3	-1/3

→ yukarı ve aşağının ortalamasını alalım türev alırken böylece görültüden daha az etkilenir

1	0	-1
1	0	-1
1	0	-1

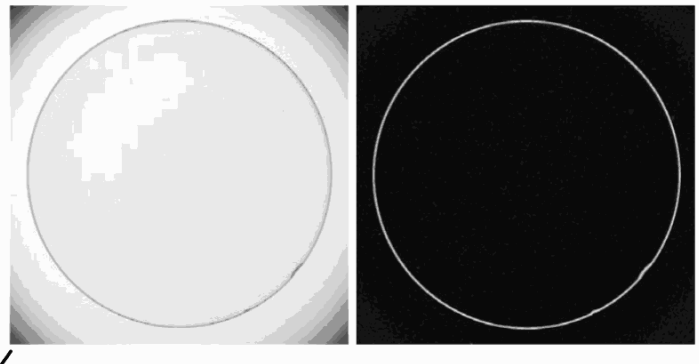
1	2	1
0	0	0
-1	-1	-1

x yönde türev alıyor

y yönünde

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1



a b

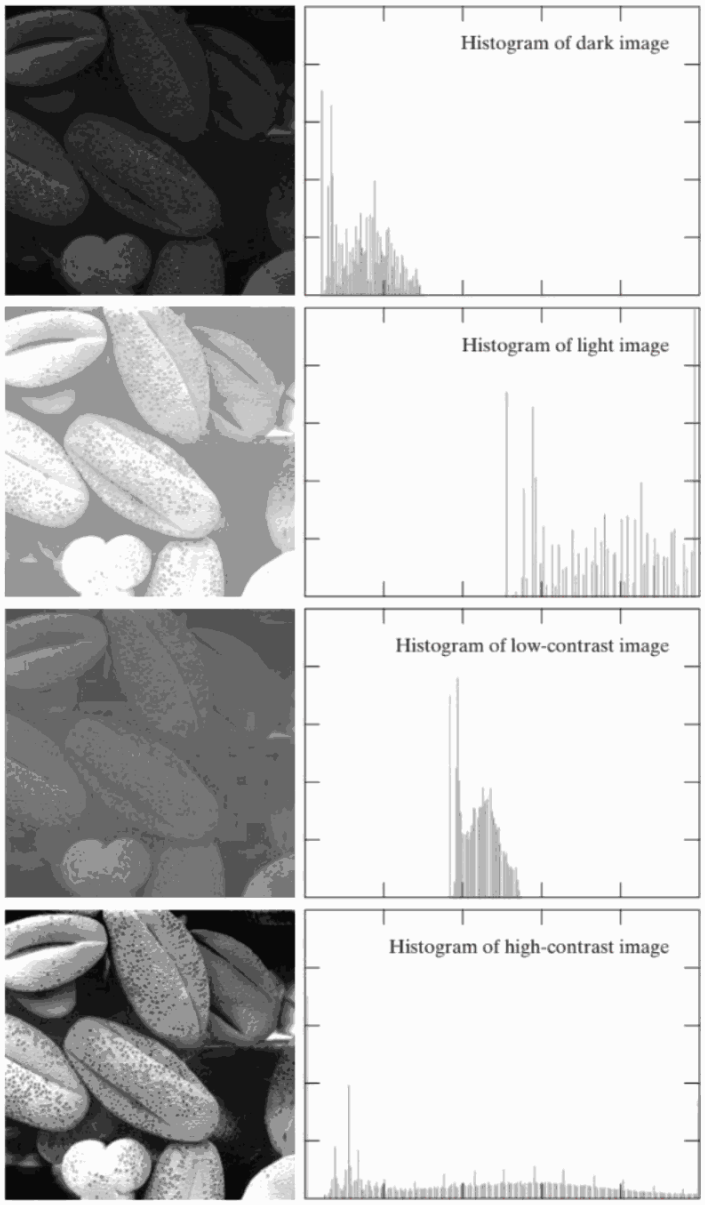
FIGURE 3.42
(a) Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Pete Sites, Perceptics Corporation.)

x ve y yönündeki sobel uygulayıp toplarsak

Gradiyent of $f(x,y)$

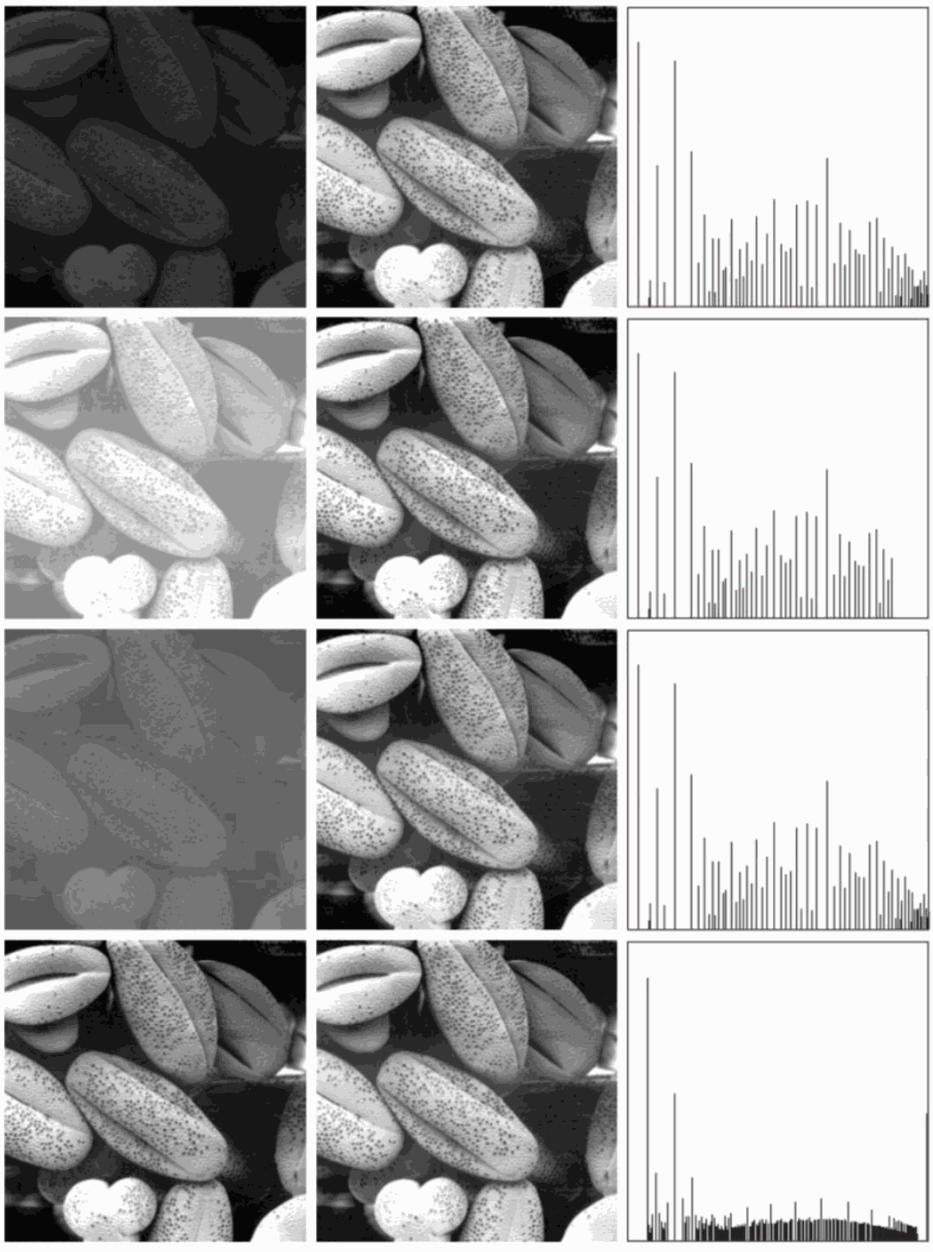
$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\text{magnitude of } \nabla f = \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 \right]^{1/2} \approx \left|\frac{\partial f}{\partial x}\right| + \left|\frac{\partial f}{\partial y}\right|$$



olasılık yoğunluk fonksiyonu eğrinin alanı 1'in altında

FIGURE 3.16 Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms.



Histogram Processing

gray levels are in $[0, L-1]$

histogram : $h(r_k) = n_k$ → piksel sayısı

$r_k = k^{th}$ graylevel 0. 1. ---255.

$$n = \sum_{k=0}^{L-1} n_k = M \times N$$

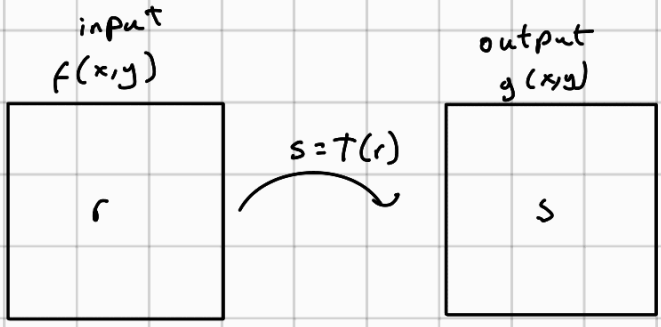
Normalize histogram $p(r_k) = n_k / n$ pdf'e benzigor

0 5 7 8 7



s

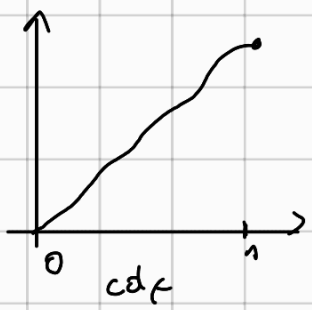
Histogram Equalization



$s = T(r)$

$$s = T(r) = \int_0^r p_r(w)\,dw$$

$$\int_0^r p(x)\,dx$$



birikimli

Ex:

0	3	3	1
1	3	1	3
0	3	3	3
1	1	1	0

r	h_r	p_r	T(r)	s	p_s
0	3	3/16	3/16	1	0
1	6	6/16	9/16	2	3/16
2	0	0	9/16	2	6/16
3	7	7/16	16/16	3	7/16

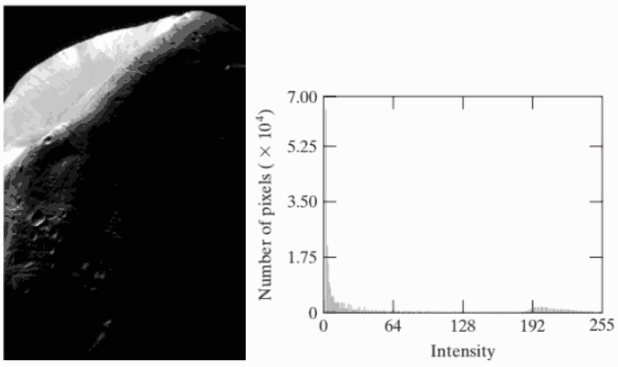
gezarken üç ile çarpalım ki 0 1 arası sıkış masın

→ uniform olmadı tam çünkü continuous değil

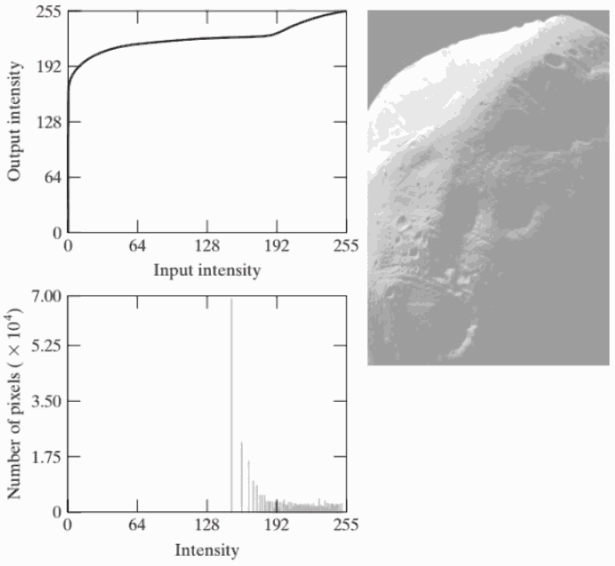
1	3		2
2	3	2	3
1	3	3	3
2	2	2	1

a b

FIGURE 3.23
(a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor.* (b) Histogram. (Original image courtesy of NASA.)



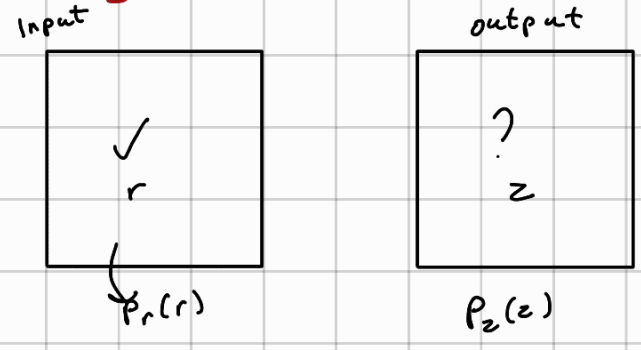
www.EBooksWorld.ir

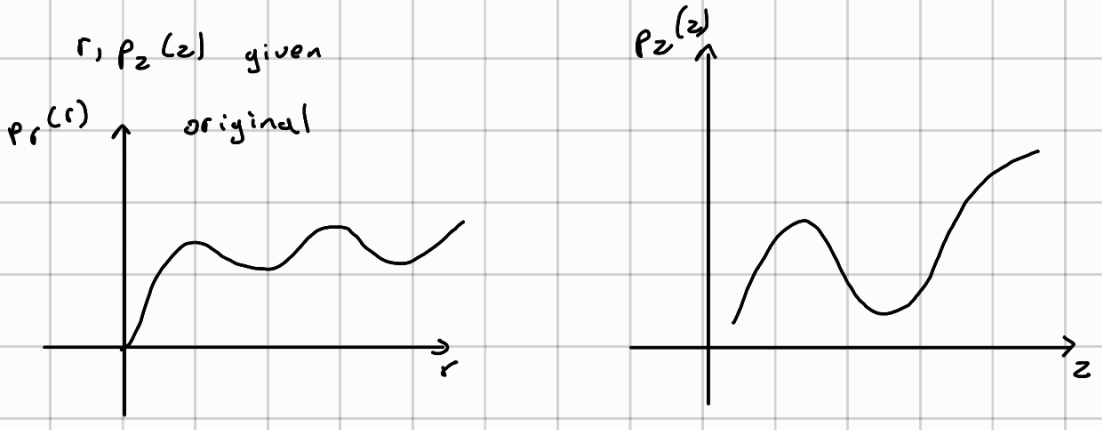


a b
c

FIGURE 3.24
(a) Transformation function for histogram equalization. (b) Histogram-equalized image (note the washed-out appearance). (c) Histogram of (b).

Histogram Matching





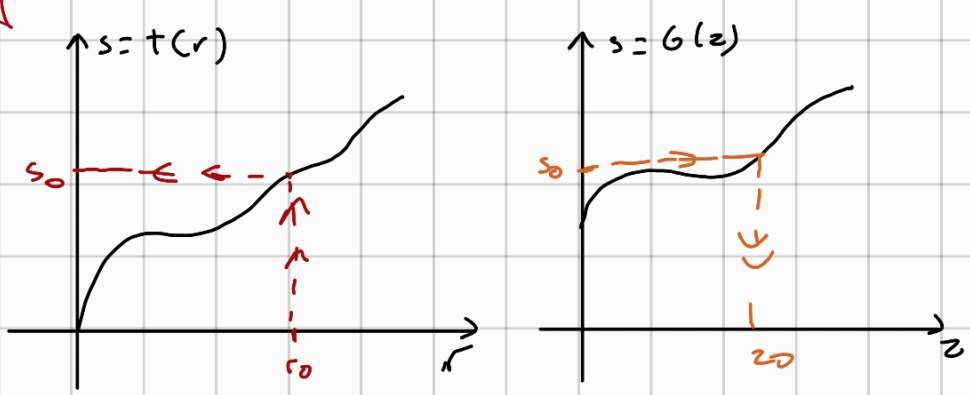
$$s = T(r) = \int_0^r p_r(w)\,dw \qquad s = G(z) = \int_0^z p_z(w)\,dw$$

ikisi aynı değil ım

$$s = G(z) = T(r)$$

$$G(z) = T(r)$$

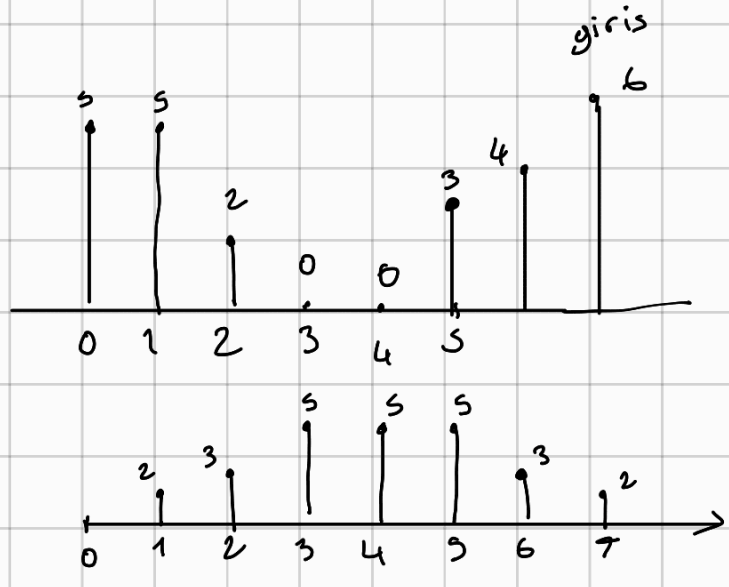
$$z = G^{-1}(T(r))$$



$$\equiv G^{-1}(T(r))$$

Input

0	1	1	0	0
1	1	0	0	1
2	2	5	6	7
5	5	7	6	6
7	6	7	7	7



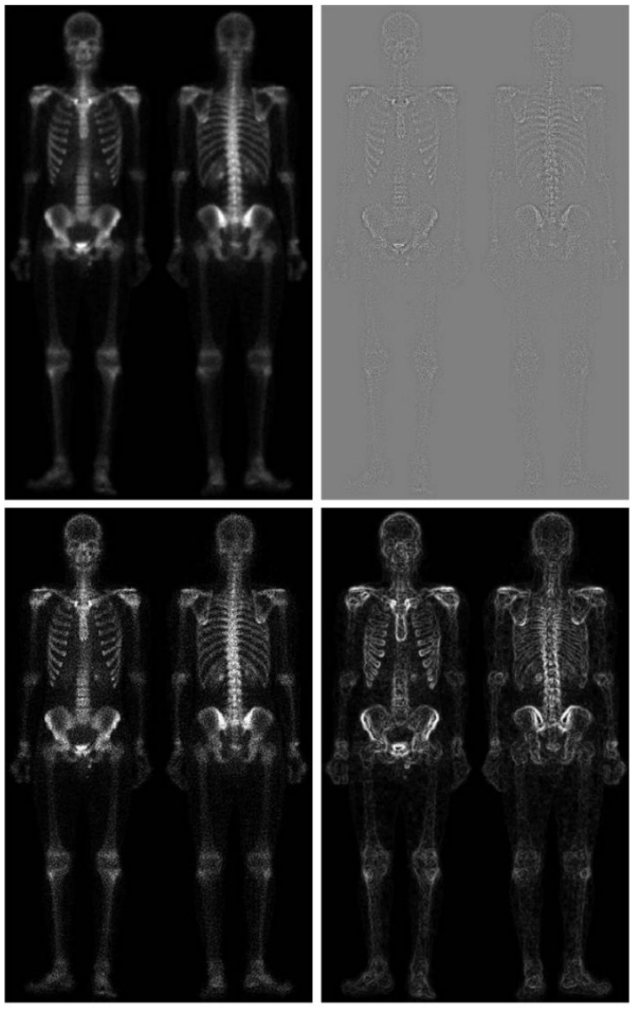
r	h(r)	Pr	T(r)	G(z)	P_z	z
0	5	5/25	5/25	0/25	0/25	2
1	5	5/25	10/25	2/25	2/25	3
2	2	2/25	12/25	5/25	3/25	3
3	0	0/25	12/25	10/25	5/25	3
4	0	0/25	12/25	15/25	5/25	3
5	3	3/25	15/25	20/25	5/25	4
6	4	4/25	19/25	23/25	3/25	5
7	6	6/25	25/25	25/25	2/25	7

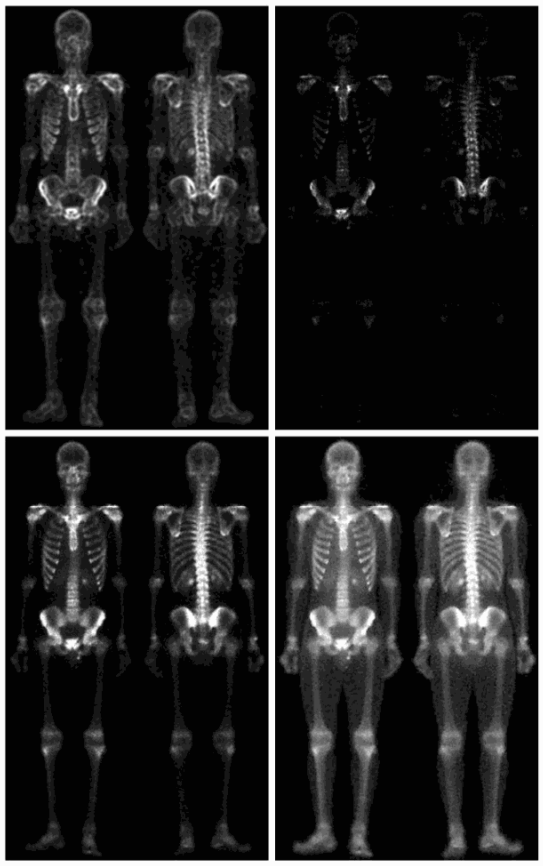
Output

2	3	3	2	2
3	3	2	2	3
3	3	4	5	7
4	4	7	5	5
7	5	7	7	7

a b
c d

FIGURE 3.43
(a) Image of whole body bone scan.
(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).





e f
g h

FIGURE 3.43
(Continued)
(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)